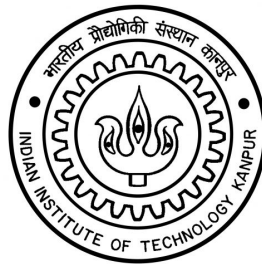


MTH 210 Course Project

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Assigned Data File: 38



Guidance of Mr Debasis Kundu

MTH210 - Statistical Computing

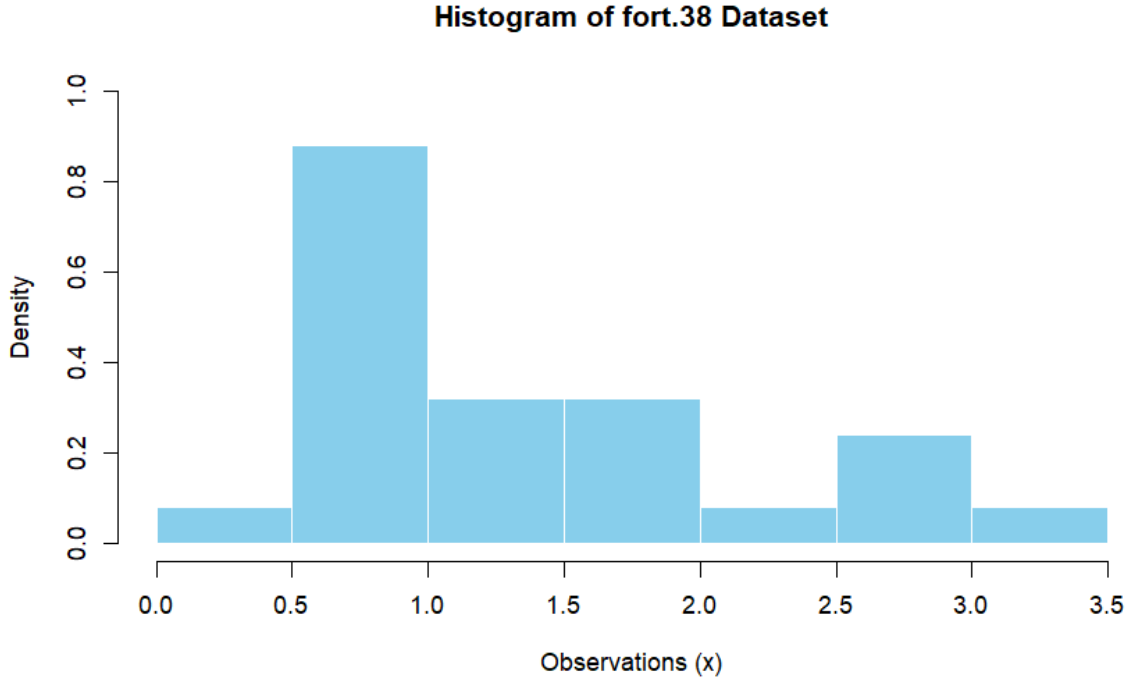
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1 Introduction and Data Summary

This report details the estimation of parameters for an Exponentiated Exponential distribution using the dataset provided in file number **38**. The dataset contains $n = 25$ independent observations. The data is assumed to be generated from the following probability density function (PDF) where $\alpha > 0$ and $\lambda > 0$:

$$f(x; \alpha, \lambda) = \alpha \lambda e^{-\lambda x} (1 - e^{-\lambda x})^{\alpha-1}$$

The primary objective is to find the Maximum Likelihood Estimators (MLEs) for the shape parameter α and the scale parameter λ using the Newton-Raphson (NR) algorithm, and to construct 95% confidence intervals using both parametric and non-parametric bootstrap methods.



2 Maximum Likelihood Estimation

To apply the Newton-Raphson method, we first derive the log-likelihood function, its gradient (first derivatives), and its Hessian matrix (second derivatives).

2.1 Log-Likelihood Function

For a sample of n independent observations $x_1, x_2, \dots, x_n$, the log-likelihood function $\ell(\alpha, \lambda)$ is given by:

$$\ell(\alpha, \lambda) = n \ln \alpha + n \ln \lambda - \lambda \sum_{i=1}^n x_i + (\alpha - 1) \sum_{i=1}^n \ln(1 - e^{-\lambda x_i})$$

2.2 Gradient Vector

Differentiating the log-likelihood function with respect to α and λ yields the following likelihood equations:

$$\frac{\partial \ell}{\partial \alpha} = \frac{n}{\alpha} + \sum_{i=1}^n \ln(1 - e^{-\lambda x_i})$$

$$\frac{\partial \ell}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i + (\alpha - 1) \sum_{i=1}^n \frac{x_i e^{-\lambda x_i}}{1 - e^{-\lambda x_i}}$$

2.3 Hessian Matrix

The second-order partial derivatives required for the Newton-Raphson update step are:

$$\begin{aligned} \frac{\partial^2 \ell}{\partial \alpha^2} &= -\frac{n}{\alpha^2} \\ \frac{\partial^2 \ell}{\partial \alpha \partial \lambda} &= \sum_{i=1}^n \frac{x_i e^{-\lambda x_i}}{1 - e^{-\lambda x_i}} \\ \frac{\partial^2 \ell}{\partial \lambda^2} &= -\frac{n}{\lambda^2} - (\alpha - 1) \sum_{i=1}^n \frac{x_i^2 e^{-\lambda x_i}}{(1 - e^{-\lambda x_i})^2} \end{aligned}$$

3 Newton-Raphson Algorithm Implementation

The Newton-Raphson method was used to iteratively solve the nonlinear system of equations $\nabla \ell = 0$. The parameter vector at the k -th iteration, $\theta^{(k)} = [\alpha^{(k)}, \lambda^{(k)}]^T$, was updated using the formula:

$$\theta^{(k+1)} = \theta^{(k)} - H^{-1}(\theta^{(k)}) \nabla \ell(\theta^{(k)})$$

Where H is the Hessian matrix and $\nabla \ell$ is the gradient vector.

Algorithm Specifications:

- **Initial Guess Values:** The initial guesses were set to $\alpha^{(0)} = 1.0$ and $\lambda^{(0)} = 0.7$. The value for λ was chosen based on the inverse of the sample mean ($1/\bar{x} \approx 1/1.425 \approx 0.7$), which is the MLE for a standard exponential distribution (where $\alpha = 1$).
- **Stopping Criterion:** The algorithm was programmed to terminate when the Euclidean norm of the difference between successive parameter estimates fell below a specified tolerance. Specifically, the criterion used was $\|\theta^{(k+1)} - \theta^{(k)}\| < 10^{-6}$.
- **Convergence:** The Newton-Raphson algorithm successfully converged in **8** iterations.

Calculated MLEs:

- $\hat{\alpha} = 4.4883$
- $\hat{\lambda} = 1.6146$

4 Bootstrap Confidence Intervals

To assess the uncertainty of the estimated parameters, 95% confidence intervals were calculated using two distinct bootstrap methodologies. For both methods, **1000 replications** were utilized.

4.1 Parametric Bootstrap

In the parametric bootstrap approach, the estimated MLEs ($\hat{\alpha}$ and $\hat{\lambda}$) were treated as the true population parameters.

1. 1000 independent samples of size $n = 25$ were generated from the $f(x; \hat{\alpha}, \hat{\lambda})$ distribution.

2. For each generated sample, the Newton-Raphson algorithm was applied to calculate a new set of bootstrap MLEs, denoted as $(\hat{\alpha}_i^*, \hat{\lambda}_i^*)$ for $i = 1, \dots, 1000$.
3. The 95% confidence intervals were constructed by extracting the 2.5th and 97.5th percentiles from the sorted bootstrap estimates.

Parametric 95% CI Results:

- CI for α : **2.6949** to **13.2841**
- CI for λ : **1.1876** to **2.5720**

4.2 Non-Parametric Bootstrap

The non-parametric approach does not assume the data follows the specific Exponentiated Exponential distribution for the resampling phase.

1. 1000 new samples of size $n = 25$ were created by resampling the original 25 data points *with replacement*.
2. The MLEs $(\hat{\alpha}_i^*, \hat{\lambda}_i^*)$ were calculated for each resampled dataset using the Newton-Raphson algorithm.
3. The 95% confidence intervals were determined by finding the 2.5th and 97.5th percentiles of these estimates.

Non-Parametric 95% CI Results:

- CI for α : **3.1129** to **9.5000**
- CI for λ : **1.1798** to **2.6296**

5 Conclusion

The Newton-Raphson algorithm proved to be highly efficient in finding the roots of the likelihood equations for this dataset. The initial guesses based on standard exponential properties provided a stable starting point, leading to rapid convergence. Both parametric and non-parametric bootstrap methods yielded comparable confidence intervals, indicating the robustness of the maximum likelihood estimates for this sample.

The parametric bootstrap assumes the underlying data perfectly follows the Exponentiated Exponential distribution, while the non-parametric method is more robust to slight model misspecifications because it relies purely on the empirical distribution of the 25 data points.